Strengthened Residual Graph and Multiscale Gated Guided Convolutional Fusion Network for Hyperspectral Change Detection

Shufang Xu, *Member, IEEE*, Xiangfei Xia, Haiwei Li, Yiyan Zhang, *Graduate Student Member, IEEE*, Runhua Sheng, *Graduate Student Member, IEEE*, Hongmin Gao, *Senior Member, IEEE*, and Bing Zhang, *Fellow, IEEE*

***Abstract*— Hyperspectral image (HSI) change detection (CD) focuses on identifying changes in the internal components of land cover and land use. Convolutional neural networks (CNNs) have made significant progress in HSI-CD. Concurrently, graph convolutional networks (GCNs) have gained considerable attention for their ability to utilize unlabeled data and explicitly exploit correlations between adjacent parcels. However, CNNs are constrained by fixed, small-size convolutional kernels, which severely limit their receptive field. On the other hand, GCNs use superpixels to reduce the number of nodes, which will lead to losing pixel-level features, resulting in partial feature representations from both networks. To leverage the strengths of both CNNs and GCNs, a model was proposed that incorporates two subnetworks: decomposed multiscale gated guided CNNs and strengthened residual graph convolution. The decomposed multiscale gated guided CNNs are designed to capture pixel-level features at various scales using different kernel sizes. A gated change information fusion (GCF) unit integrates these multiscale pixel-level features. Meanwhile, the strengthened residual graph convolution was used to aggregate change information, which can prevent node information from becoming homogeneous. Additionally, a feature fusion module (FFM) is employed to combine features from the two subnetworks. The proposed model effectively utilizes both multiscale convolution and graph features, facilitating the learning of multilevel contextual semantic features. The experimental results on three HSI datasets demonstrate that this model outperforms several state-of-the-art methods. The code is available at https://github.com/zhangyiyan001/srgmgn.**

***Index Terms*— Change detection (CD), convolutional neural network (CNN), graph convolutional network (GCN), hyperspectral image (HSI), superpixel.**

Received 12 September 2024; revised 5 November 2024; accepted 25 November 2024. Date of publication 27 November 2024; date of current version 9 December 2024. This work was supported in part by the Open Research Fund of Shaanxi Key Laboratory of Optical Remote Sensing and Intelligent Information Processing under Grant KF20230301 and in part by the Headquarters of the State Grid Corporation of China under Grant 5108-202340062A-1-1-ZN. *(Corresponding author: Hongmin Gao.)*

Shufang Xu is with the College of Computer Science and Software Engineering, Hohai University, Nanjing 211100, China, and also with Shaanxi Key Laboratory of Optical Remote Sensing and Intelligent Information Processing, Xi'an 710119, China (e-mail: xushufang@hhu.edu.cn).

Xiangfei Xia, Runhua Sheng, and Hongmin Gao are with the College of Computer Science and Software Engineering, Hohai University, Nanjing 211100, China (e-mail: 231607010088@hhu.edu.cn; 221307040017@hhu.edu.cn; gaohongmin@hhu.edu.cn).

Haiwei Li is with the Key Laboratory of Spectral Imaging Technology of CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China (e-mail: lihaiwei@opt.ac.cn).

Yiyan Zhang is with the College of Information Science and Engineering, Hohai University, Changzhou 213200, China (e-mail: zhangyiyan@hhu.edu.cn).

Bing Zhang is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: zb@radi.ac.cn).

Digital Object Identifier 10.1109/TGRS.2024.3508063

I. Introduction

With the continuous improvement of spectral imaging technology, hyperspectral images (HSIs) are increasingly utilized in Earth observation [1]. Unlike other remote sensing images, HSIs are distinguished by their meticulous spectral sampling across an extensive range of wavelengths, which can facilitate the acquisition of continuous spectral data. This elevated spectral resolution empowers HSIs to effectively document intricate land cover dynamics.

And HSI is still a relatively new technology in biomedical applications, but it has a wide range of potential uses as a noninvasive method for diagnosing and evaluating treatments. This is because HSIs have the ability to extract the spectral signature of each pixel from an HSI while providing information about the composition of different tissues and their spatial distribution when measuring the reflection and absorption of light at different wavelengths [2], [3]. At specific wavelengths, the chemical composition and physical characteristics of tissues in different pathological states have different reflectances, absorbances, and electromagnetic energies, which are manifested by differences in characteristic optical spectral peaks. By analyzing these spectral signals, qualitative or quantitative detection of tissue state information can be realized, and according to the spatial distribution information provided by high-spectral images, different pathologies of tissues can be visualized, so as to diagnose tissue disease states. In the medical field, the capabilities of HSIs are increasingly being used in disease detection and surgical guidance. The literature in the past decade shows that as a new noncontact optical diagnostic technology, HSI provides an effective auxiliary diagnosis method for clinical medicine through spectral image information and has great development potential [4], [5].

Change detection (CD), a key application of remote sensing, benefits from HSIs' rapid processing, extensive coverage, spectral diversity, and periodicity [6], [7]. By analyzing HSIs from different times at the same location, HSI-CD effectively monitors dynamic changes and spatial differences on the land surface. Its applications in agriculture, forestry [8], [9], land

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.

cover renewal [10], [11], and natural disaster assessment [12], [13] have made it a focal point in remote sensing research in recent years [14], [15], [16], [17].

Traditional hyperspectral (HS) CD methods generally follow a unified framework and can be achieved in three steps.

1) Image preprocessing, geometric correction, radiometric correction, denoising, and other operations on the initial image to reduce the pseudo-change information caused by the difference in image representation, so that the image is more suitable for comparison to determine the difference.
2) Differential image generation utilizes pixel-level algebra, feature-level transformation, and object-level classification methods to create change maps.
3) Outcome evaluation, in which the change map is compared to the visually interpreted ground truth (GT), and the results of the CD are evaluated using some appropriate metrics in order to easily compare the various CD methods.

In recent years, many traditional CD methods have focused on differential image generation, and these methods mainly include the following four categories: 1) algebra-based methods (such as image difference [18] and image rationing [19]); 2) transform-based approaches (such as change vector analysis (CVA) [20], principal component analysis (PCA) [21]); 3) classification-based approach [22]; and 4) other methods (Markov model [23] and fuzzy clustering [24]).

However, traditional methods exhibit several drawbacks. Algebraic methods suffer from high dependence on band selection and prediction, and low detection accuracy for multiple variation types, and have difficulty in determining appropriate thresholds. Transformation methods face issues such as loss of spectral information and significant computational effort required to generate new components, especially when the change area is large. Classification methods are limited by their heavy reliance on the performance of the classifier, often neglect the correlation between bitemporal HSIs, and are influenced by the choice of classification algorithm, which affects their robustness. Additionally, demixing methods are hindered by their dependence on the accuracy of spectral decomposition and are often tailored to specific problems or assumptions, leading to limited versatility.

In recent years, deep learning has gained significant attention from researchers across various fields because of its powerful learning and representation capabilities. It has provided new solutions for intelligent processing of remote sensing images and demonstrated great potential in CD, driven by advancements in computing power and the availability of large volumes of remote sensing data. One of the reasons why deep learning-based methods can achieve high detection accuracy is that this type of feature does not rely on manual feature extraction experience.

As a typical deep learning architecture, the convolutional neural network (CNN) can efficiently abstract concrete image data into high-level features, showing excellent detection performance and unlimited potential. For example, Wang et al. [25] implemented HSI-CD in an end-to-end manner by integrating multiple subpixel information sources into a hybrid affinity matrix. And in order to combine spectral and spatial features and transform the CD problem into a similarity measurement problem between two tensors, Zhan et al. [26] proposed a spectral space CNN with a Siamese (SSCNN-S) architecture. Furthermore, Ou et al. [27] introduced a CNN framework called low–fast band selection-feature fusion grouping network (SFBS-FFGNET) that employs slow–fast band selection to reduce the number of bands and mitigate the Hughes phenomenon, while enhancing feature fusion through grouping. Luo et al. [28] proposed a multiscale differential change feature fusion network (MSDFFN), which designed to capture subtle changes at various scales. Additionally, long short term memory (LSTM) networks are frequently used in HSI-CD to exploit temporal correlations between multiphase images. According to [29], a multipath convolutional LSTM (MP-ConvLSTM) neural network utilizes CNN and LSTM to extract multiscale spatiotemporal spectral features in parallel. Meanwhile, Qu et al. [30] explored the role of multilevel fusion strategies in HSI-CD by proposing a multilevel encoder–decoder attention network (ML-EDAN), which analyzes multilevel spatiotemporal–spectral correlation information using LSTM. Furthermore, Gao et al. [31], [32], [33], [34], [35], Su et al. [36], Rao et al. [37], and Sun et al. [38], [39] conduct many research on HSIs.

Although the above CNN-based methods have achieved satisfactory detection results, they are usually limited by the following factors: first, the traditional CNN uses HS plaques as the detection unit, so that feature extraction is carried out in fixed plaques. This not only leads to the loss of edge information and limits the discriminative ability of the network, but also hinders the exploration of long-range correlation [40], [41]. And to solve this problem, some studies use dilated and deformed convolutions to extend the received information. Specifically, Song et al. [42] expanded the receptive field of the network by employing dilated convolutions with varying dilation ratios instead of traditional convolutions. Yang et al. [43] utilized continuously expanding deformation feature extraction blocks to flexibly obtain subpixel information beyond the sampling locations. Recently, researchers in HSI processing have explored Transformer architectures to address the receptive field limitations of CNN-based methods [44], [45], [46]. For instance, Wang et al. [46] proposed the spectrum-space-time-former (SST-Former), which includes a spectral Transformer and a spatial Transformer to extract spectral and spatial information, respectively. The temporal Transformer is then used to learn change-related features and derive change rules between bitemporal HSI objects. The self-attention mechanisms in these Transformer-based methods enable them to capture features over a larger receptive field. Zhang et al. [47] propose a novel cross-modal feature aggregation and enhancement network for the joint classification of HSI and light detection and ranging (LiDAR) data. Zhang et al. [48] also present a new model called multimodal transformer network (MTNet) that leverages transformer advantages to capture both the specific and shared characteristics of HS and LiDAR data. However, these methods often require extensive training and testing time due to high computational demands and slow convergence. Moreover, Tu et al. [49], [50], [51], [52], Liao et al. [53], and He et al. [54], [55] have made many contributions to HSI processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.

Recently, graph neural networks (GNNs), which specialize in extracting features from irregular topological structures, have gained significant attention across various research domains [56], [57]. Convolutional GNNs (ConvGNNs) extend traditional convolution operations from fixed meshes to topological graph structures, and the ConvGNNs create node representations by aggregating features from both the node itself and its neighbors [58]. Several studies have recently incorporated graph convolutional networks (GCNs) into HSI processing [40] due to their robust ability to model long-range spatial relationships. For instance, Qu et al. [59] utilized GCNs for HS CD (HSI-CD) to effectively extract and amplify differential features. Additionally, Wang et al. [60] introduced a novel module based on graph attention networks (GATs) to exploit the spatiotemporal joint correlations of multitemporal HSIs.

Although many HSI-CD methods have been proposed, they still face the following challenges. First, existing frameworks for HSIs feature extraction often struggle in complex topological spaces, leading to inaccuracies in capturing change information between multitemporal images. Second, the high dimensionality of HSIs poses challenges in forming spatial pixel-level graph structures, and single convolutional layers are often inadequate for capturing detailed information. Additionally, using superpixel graph structures can hinder the detection of fine-grained changes. Third, with limited labeled samples, effectively leveraging information from unlabeled samples to address the deficiencies of labeled data remains a significant challenge. To address these issues, a CD algorithm based on strengthened residual graph convolution and decomposed multiscale gated guided convolutional fusion (SRGMGN) is proposed. This algorithm integrates pixel-level variation features, guided by the multiscale gated guided convolutional module, with the original bitemporal superpixel features, thereby reducing the network's reliance on extensive training samples. The main contributions of this work are as follows.

1) By integrating decomposed multi-scale gated guided CNNs module (MSGM) with strengthened residual graph convolution module (SRGM) in the SRGMGN framework, the network effectively captures both subtle and large-scale variations, demonstrating significant potential for low-sample-rate CD tasks.
2) To address the issue of consistent node information and diminished change information in traditional GCNs, a strengthened residual connection method was proposed to enhance the expression of superpixel-level features and improve network robustness.
3) To facilitate the deep fusion of superpixel and pixel features, a feature fusion module (FFM) module was proposed to enhance feature integration and extracts critical identifying information for more accurate CD results.
4) A hybrid enhanced feature extraction framework SRGMGN based on SRGM, MSGM, and FFM was proposed. SRGMGN has strong multilevel feature capture capabilities. Through in-depth experiments on three datasets, the effectiveness of the model is verified.

The remainder of this article is organized as follows. Section II analyzes the work, including the application of GNNs to HSI processing. Section III introduces the proposed HSI-CD method, SRGMGN, including the superpixel segmentation module, two important branches, feature fusion, and CD. Section IV introduces the experimental results and visual analysis of this article, including plain sampling rate experiment, low sampling rate experiment, and ablation experiment. Conclusions are provided in Section V.

II. Related Work

GNNs have attracted great interest because of their ability to model topological data and play an important role in solving topological problems such as node classification, motion recognition, and traffic prediction. GNNs were first proposed by Scarselli et al. [61], [62] as a semi-supervised neural network architecture that can directly process graph-structured data. Influenced by the great success of CNNs in remote sensing image processing, people have begun to work on redefining convolution operations acting on graph data and calling them ConvGNNs. ConvGNN is designed to extend convolution operations to graph-structured data to efficiently extract topological features. Depending on the working domain, ConvGNN can be divided into two categories: frequency domain-based methods and spatial domain-based methods. GCNs are typical spectrum-based ConvGNNs. The network transfers the convolution operation of the traditional Euclidean space to the graph network and uses the Laplace matrix of the graph to generate the convolution operation of the graph. However, due to the high computational complexity of eigenvalue decomposition of the Laplace matrix, this network architecture is usually only suitable for the calculation of small-scale graphs. In the field of HSI-CD, as a new framework for directly learning graph structure data, GNN has achieved great success in the field of computer vision by formulating certain strategies for the nodes and edges of graph to realize the standardized representation of graph structure data. While GNN has attracted a great deal of attention in the field of HSI processing in recent years, its applications are still scarce. In particular, the application of GNN in HSI-CD is still in its infancy, and many issues need to be further studied.

III. Proposed Method

A. Overview

In this section, the process of the simple linear iterative clustering (SLIC) method was first introduced and then described the overall structure of the proposed method in detail, which includes graph learning, MSGM module, SRGM module, FFM module, the gated change information fusion (GCF) unit, and the loss function.

Define the bitemporal HSIs as $\mathbf{X}_1, \mathbf{X}_2 \in \mathbb{R}^{H\times W\times B}$, and $\mathbf{V} \in \mathbb{R}^{H\times W}$ is the corresponding GT, where H and W denote the height and width of the spatial dimension of HSIs, respectively, and B denotes the number of bands of HSIs. The proposed SRGMGN is a hybrid neural network that includes a superpixel and pixel feature extraction module and feature fusion module; the overall framework is shown in Fig. 1. It is mainly composed of two parts: the S-branch and the P-branch. In addition, some preprocessing work was carried out, which included the superpixel segmentation. In the superpixel segmentation module, the SLIC method is used

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.

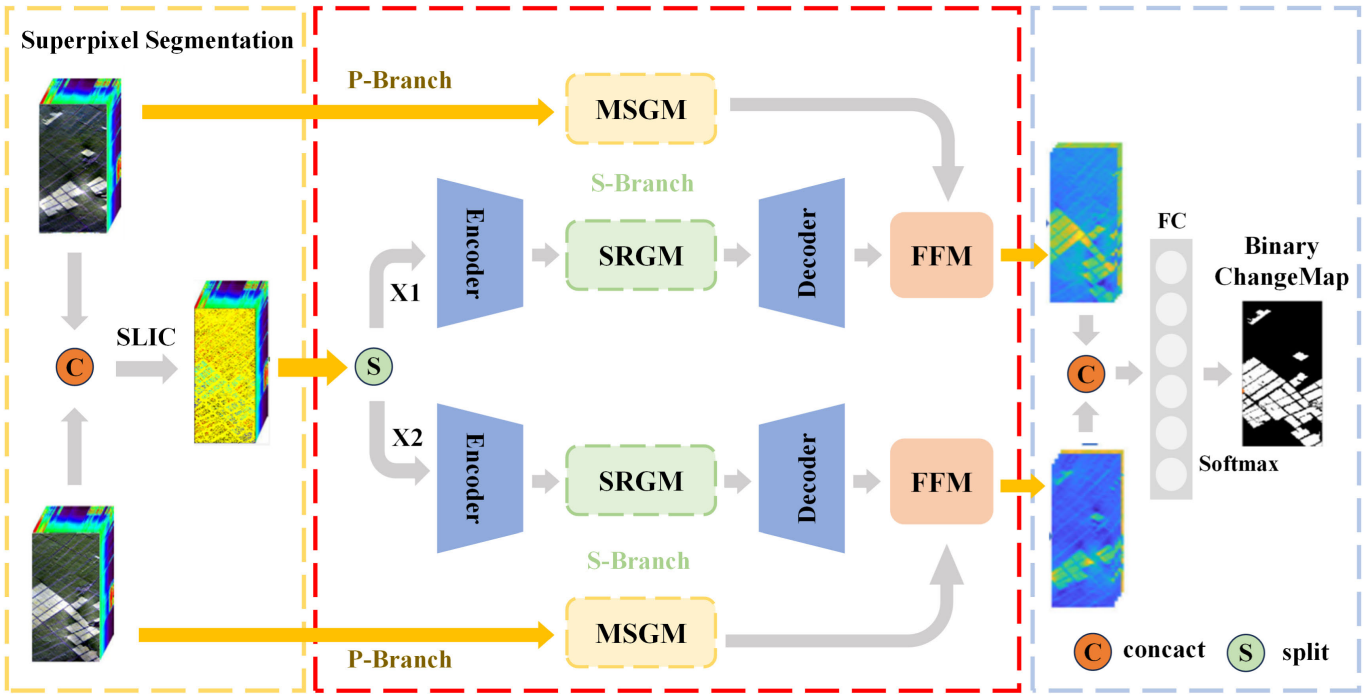


Fig. 1. Overall architecture of the proposed method SRGMGN.

to complete the segmentation of HSI. This module generates correlation matrices and adjacency matrices, which will be used by the encoder and decoder. In the S-branch, the two HSIs are first fed into the encoder and converted into a matrix of nodes. Then, they are input into the proposed SRGM module. The proposed SRGM algorithm can complete the propagation of features in the topology and obtain the feature matrix of the graph. Finally, the decoder divides the graph feature matrix and converts it into superpixel-level features. Meanwhile, the P-branch generates pixel-level features through the MSGM module and fuses them with superpixel-level features through the proposed FFM module to complete the HSI-CD task. The specific implementation steps of the model are given in Algorithm 1.

Algorithm 1 Algorithm of SRGMGN

Input: Hyperspectral images X_1, X_2; GT map Y; learning rate lr; training epochs E.
Output: Predict change map, $\hat{Y}$.

```
Segment X1, X2 into superpixels via SLIC algorithm.
for e = 1 to E do
    Construct incidence matrix Q by Eq. (2).
    Encode X1, X2 by Eq. (3).
    Implement graph convolution to X1, X2 by Eq. (5)~(11), noted as G1, G2.
    Decode G1, G2 by Eq. (4), noted as G3, G4.
    Implement convolution to X1, X2 by Eq. (12)~(14), noted as C1, C2.
    Input G1 and C1 into FFM module by Eq. (15)~(18), noted as F1.
    Input G2 and C2 into FFM module by Eq. (15)~(18), noted as F2.
    Concatenate F1 and F2, noted as F
    Predict the change map.
    Ŷ = Softmax(F)
    Calculate the loss L between Y and Ŷ by Eq. (19).
    Back-propagate L and solve it with Adam optimizer.
end for
```

B. Superpixel Segmentation

HSI typically contains a large number of pixels in the spatial dimension. In constructing a graph structure, if each pixel of HSI is divided into a graph node, it will cause a huge amount of computation. To address this issue, this article adopts the superpixel segmentation algorithm SLIC method, which divides all the pixels into superpixels according to spatial similarity. In the process of superpixel segmentation, the scale of segmentation is denoted as c, which means the average number of pixels per superpixel, and thus, the number of superpixels is $K = N/c$, where $N = H\times W$ denotes the number of pixels in each band of HSI. The operation procedure of the SLIC method is as follows: first, in order to ensure that the input HSIs after the superpixel segmentation operation keep the same structure, the two input multitemporal HSIs are cascaded along the channel dimension, and then, superpixel segmentation is performed on the cascaded image to obtain the segmentation result $\widehat{\mathbf{X}_3} \in \mathbb{R}^{N\times 2B}$, which is defined as

$$\widehat{\mathbf{X}_3} = \mathrm{SLIC}([\mathbf{X}_1 \| \mathbf{X}_2]) \tag{1}$$

where $\mathbf{X}_1 \in \mathbb{R}^{H\times W\times B}$ and $\mathbf{X}_2 \in \mathbb{R}^{H\times W\times B}$ represent two input multitemporal HSIs, H and W denote the height and width of images, respectively, B is the number of bands of each image, $\mathbf{X}_1 \| \mathbf{X}_2$ represents the concatenation operation, and SLIC denotes the superpixel segmentation operation. Then, to determine the dual-branch graph structure, $\widehat{\mathbf{X}_3}$ is then split into $\widehat{\mathbf{X}}_1 \in \mathbb{R}^{N\times B}$ and $\widehat{\mathbf{X}}_2 \in \mathbb{R}^{N\times B}$ along the channel dimension. See Fig. 2 for the process of the SLIC method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.

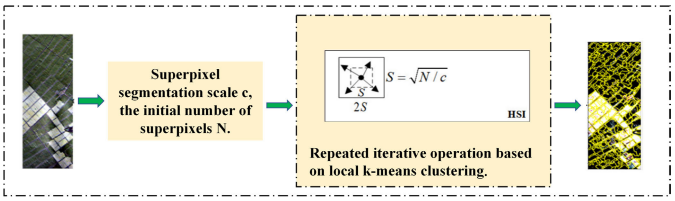


Fig. 2. Process of the SLIC method.

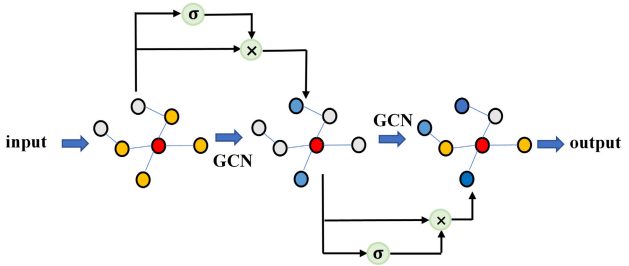


Fig. 3. Overall architecture of SRGM.

C. SRGM

In the SRGM module, the graph nodes obtained by superpixel segmentation are used to construct the graph structure of two branches. Each branch is composed of two layers of GCN with the same topological structure but different node values. At each layer, the network uses the enhanced residual connection to strengthen the difference information of each graph node, which effectively suppresses the problem of graph node information consistency. See Fig. 3 for the process of the SRGM module.

At the HSI preprocessing stage, the network constructs the graph by superpixel segmentations and obtains the transformation matrix $\boldsymbol{Q} \in \mathbb{R}^{HW\times Z}$ between pixels and superpixels, where Z denotes the number of superpixels

$$\boldsymbol{Q}_{i,j} = \begin{cases} 1, & \text{if } \hat{\boldsymbol{X}}_i \in \boldsymbol{S}_j \\ 0, & \text{if } \hat{\boldsymbol{X}}_i \notin \boldsymbol{S}_j, \end{cases} \qquad \hat{\boldsymbol{X}} = \text{Flatten}(\boldsymbol{X}) \tag{2}$$

where $\hat{X} \in \mathbb{R}^{HW\times B}$ is the shaping of the HSI original pixels, and the two spatial dimensions are flattened to combine a matrix, so $\mathcal{Q}_{i,j}$ reflects the relationships between the ith pixel element $\hat{X}_i$ and the jth superpixel element $\mathbf{S}_j$. Then, the following formulas are used to achieve the transformation between the nodes of graph and the superpixels:

$$H = \text{Encoder}(X;\, Q) = \bar{Q}^{\mathrm{T}}\, \text{Flatten}(X) \tag{3}$$

$$\tilde{X} = \text{Decoder}(H;\, Q) = \text{Reshape}(QH) \tag{4}$$

where $\bar{\boldsymbol{Q}}$ denotes the column-normalized $\mathcal{Q}$, H represents a node composed of superpixels, and Reshape$(\cdot)$ represents the spatial dimension of the restored flattened data. Encoder$(\cdot)$ means convert the pixmap to a graph node, and Decoder$(\cdot)$ represents the mapping of superpixel nodes to pixels.

The construction of the graph model in the GCN depends heavily on the establishment of an undirected graph, which is used to describe the set of nodes, edges of the graph structure, and the adjacency matrix, which is composed of the similarity weights of the edges between connected nodes. In this article, define the relationship of spectral features in an HSI as an undirected graph of $\bar{\mathbf{G}} = (\bar{\mathbf{V}}, \mathbf{E})$, where $\mathbf{V}$ denotes the node set with $\mathbf{V} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_N\}$ and $\mathbf{E}$ is the edge set. The adjacency matrix defined as $\mathbf{A} \in \mathbb{R}^{N\times N}$ is used to describe the internal association between nodes. The element $\mathbf{A}_{i,j}$ in $\mathbf{A}$ represents the weight of the edge between node $\mathbf{v}_i$ and node $\mathbf{v}_j$, which is defined as follows:

$$\mathbf{A}_{i,j} = \exp\left(-\frac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2}{\sigma^2}\right) \tag{5}$$

where hyperparameter σ is empirically set to 0.1 in the experiments. The vectors $\mathbf{x}_i$ and $\mathbf{x}_j$, respectively, represent the corresponding spectral features of graph node $\mathbf{v}_i$ and node $\mathbf{v}_i$, which are determined by the superpixel segmentation. Furthermore, each small region divided by SLIC is a graph node $\mathbf{v}_i$ in the graph $\mathbf{G}$, and the average value of the spectral vector of the pixels contained in the small region (i.e., superpixel) is the feature value of the graph node (i.e., $\mathbf{x}_i$). The closer $\mathbf{A}_{i,j}$ is to 1, the more similar the spectral features of nodes $\mathbf{v}_i$ and $\mathbf{v}_j$ are. This article considers that there is a direct connection between any two graph nodes in graph G, which helps to obtain the global information of the image. With the construction of node features and adjacency matrix, the initial graph structures of two multitemporal HSIs in the GCN are successfully established.

For a graph signal $x \in \mathbb{R}$ consisting of graph node eigenvectors, give a diagonal matrix filter $g_\theta = \mathbf{diag}(\theta)$, where $\theta \in \mathbb{R}^N$ are learnable parameters in the Fourier domain. Then, the graph convolution operator can be written as follows:

$$g_\theta \star x = U g_\theta U^T x \tag{6}$$

where $\boldsymbol{U}$ is the matrix of eigenvectors of $\boldsymbol{L}$, i.e., $L = U\Lambda U^{\mathrm{T}}$, and Λ is the diagonal matrix of eigenvalues of $\boldsymbol{L}$.

It should be noted that the direct convolution calculation requires the Laplacian eigenvector, which is too computationally complex to implement for large graphs. To circumvent this obstacle, one possible way is to approximate the filter to the order of k Chebyshev polynomials, such as K-localized convolution

$$g_\theta \star x \approx \sum_{k=0}^{K} \theta'_k T_k(\tilde{\boldsymbol{\Lambda}}) x \tag{7}$$

where $\tilde{\Lambda}^2 = -(2/\lambda_{\max})\boldsymbol{\Lambda}^2 - \boldsymbol{I}$ with $\lambda_{\max}$ being the largest eigenvalue of $\boldsymbol{L}$. The Chebyshev polynomials are defined recursively by $T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x)$ with $T_0(x) = 1$ and $T_1(x) = 1$.

Moreover, this article considers the first-order approximation of the Chebyshev polynomial (i.e., $K = 1$) and further simplifies the maximum eigenvalue $\lambda_{\max} = 2$. In this way, the formula can be rewritten as

$$g_\theta \star x \approx \theta\left(I + D^{-\frac{1}{2}} A D^{-\frac{1}{2}}\right) x. \tag{8}$$

To avoid numerical instability and gradient explosion or disappearance caused by stacking of convolutional layers $I + D^{-(1/2)}AD^{-(1/2)}$ is replaced by $\widetilde{D}^{-(1/2)}\widetilde{A}\widetilde{D}^{-(1/2)}$, where $\widetilde{A} = \boldsymbol{I} + \boldsymbol{D}, \widetilde{\boldsymbol{D}}_{ii} = \sum_j^{\cdot} \widetilde{\boldsymbol{A}}_{ij}$. As a result, the convolution operation of the GCN model can be expressed as follows:

$$\boldsymbol{H}^l = \sigma\left(\widetilde{\boldsymbol{D}}^{-\frac{1}{2}} \widetilde{A} \widetilde{\boldsymbol{D}}^{-\frac{1}{2}} \boldsymbol{H}^{l-1} \boldsymbol{W}^l\right) \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.

where H^l denotes the output of the lth layer and $\sigma(\cdot)$ represents the activation function, such as the rectified linear unit (ReLU) function used in SRGM, and the graph convolution operation is as follows:

$$\boldsymbol{H}^l = \sigma\left(\widetilde{\boldsymbol{D}}^{-\frac{1}{2}}\widetilde{\boldsymbol{A}}\widetilde{\boldsymbol{D}}^{-\frac{1}{2}}\sigma\left(\widetilde{\boldsymbol{D}}^{-\frac{1}{2}}\widetilde{\boldsymbol{A}}\widetilde{\boldsymbol{D}}^{-\frac{1}{2}}\hat{\boldsymbol{H}}^{l-1}\boldsymbol{W}_1^l\right)\boldsymbol{W}_2^l\right) \tag{10}$$

where $\hat{H}^{l-1}$ represents the normalization of input H^{l-1}, and W_1^l and W_2^l represent the weight matrix of the two hidden GCN layers, respectively.

The GCN is usually stacked and uses multiple convolutional layers with fixed kernels. For single-layer GCN, it can be seen that nodes aggregate features and update information from the nearest neighbor nodes through the adjacency matrix A at the GCN layer. However, too many layers stacked will lead to the consistency of graph node information and the reduction of change information, so the method of strengthening residual connection is adopted, which can improve the discriminative performance of features. Hence, the convolution result of each layer in SRGM can be expressed as follows:

$$H_*^l = H^l + H^{l-1} * \text{sigmoid}(H^{l-1}) \tag{11}$$

where H_*^l is the final output of each layer, and H_*^l contains important information from the previous layer.

D. MSGM

The 2-D CNN is widely used in the field of natural image processing due to its powerful spatial context feature extraction ability, and the formula is

$$T^l(x, y) = \sigma\left(\sum_{h=0}^{H-1}\sum_{w=0}^{W-1} s^l(h, w)\boldsymbol{T}^{l-1}(x+h, y+w) + \boldsymbol{b}^l\right) \tag{12}$$

where $T^l(x, y)$ represents the output data of the lth layer at the position (x, y), $T^{l-1}(x+h, y+w)$ represents the specific value of the feature map at the $(x+h, y+w)$ position in the $(l-1)$th layer, s^l represents the specific value of the convolution kernel of the l layer at the position (h, w), H and W represent the height and width of the convolution kernel, respectively, $\boldsymbol{b}^l$ represents the bias of the lth layer, and $\sigma(\cdot)$ represents the activation function; in MSGM, all use ReLU.

To enhance the feature collection capability of CNN-based networks and break through the limitation of the receptive field of a single fixed kernel, a parallel multiscale information collection network was designed, which contains kernels of various sizes. Fig. 4 shows the structure of an MSGM with three parallel paths, which were divided into three stages. The kernel sizes were $K = 7 \times 7$, 5×5, and 3×3. Then, in order to reduce the number of parameters while maintaining the same receptive field of convolution, the network decomposes the convolution kernel integral of 7×7 and 5×5 into triplicate 3×3 and twice 3×3. See Fig. 4 for the MSGM.

To reduce redundant image information and prevent overfitting, the network adds a layer of average pooling layer between the two layers of convolution and uses the statistical information of subblocks in a certain area to describe the spatial distribution of all pixels in the area. Here, this article adopts the average pooling to calculate the average value of all pixels in the area subblock. The pooling layer uses a pooling window of size 3×3, the stride of each movement is 1, and the padding size is set as 1 to keep the image size unchanged.

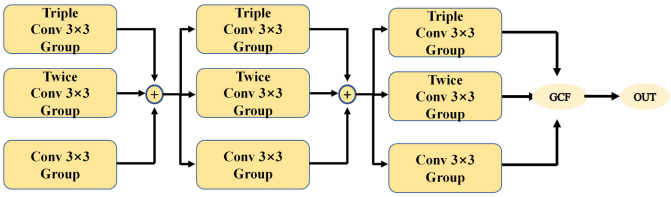


Fig. 4. Overall architecture of MSGM.

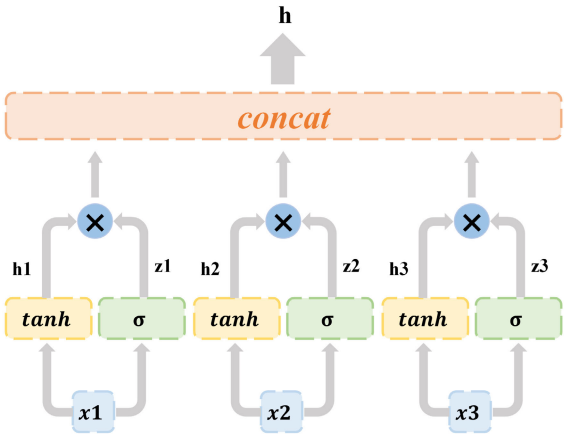


Fig. 5. Overall architecture of GCF.

In MSGM, for the input of the next convolutional layer, cross-path fusion was used to integrate the features that extracted by different scale convolution kernels. The traditional multipath multiscale CNN is relatively independent for each branch, and the features captured by each are integrated at the end. There is a gap between large-scale information and small-scale information, which limits the feature fusion ability of the network. In contrast, this article repeatedly introduces cross-path connections among the three paths, so that the convolution module that captures large-scale information and small-scale information can be fully integrated before entering the next layer, for input T^{l-1}

$$T^{l-1} = T_1^{l-1} + T_2^{l-1} + T_3^{l-1} \tag{13}$$

where $\boldsymbol{T}_1^{l-1}$, $\boldsymbol{T}_2^{l-1}$, and $\boldsymbol{T}_3^{l-1}$ represent the features from convolution kernel sizes $k = 7 \times 7$, 5×5, and 3×3, respectively.

After completing the two-stage convolution through MSGM, the GCF unit was used to splice the features, T_1, T_2, and T_3, which obtained by the three branches in the channel dimension for further fusion in subsequent modules. Fig. 5 shows the structure of a GCF unit

$$T_{\text{out}} = \text{GCF}(T_1, T_2, T_3) \tag{14}$$

where GCF denotes the gated fusion operation. The GCF is to enhance feature representation by effectively integrating information across different channels. And it can enhance feature discrimination: by fusing features at a global level, the GCF unit can highlight important characteristics and reduce noise, thereby improving the model's ability to distinguish

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.

between different classes or changes in data. Leverage global context: it captures contextual information from the entire feature map rather than relying solely on local information. This broader perspective helps the model understand relationships and dependencies between different features. Focus on important features: the GCF unit uses attention mechanisms to prioritize significant features while downplaying less relevant ones. This allows the network to concentrate on the most critical aspects of the data, improving overall performance. Facilitate multiscale fusion: by integrating features from different scales, the GCF unit supports a more comprehensive understanding of the data, allowing the model to capture both fine-grained details and broader patterns.

Use the multiplication gate that assigns importance to various features at the same time, with each x_i corresponding to a feature of a different receptive field.

Each eigenvector provides a tanh activation function for neurons; by using the tanh activation function, it has three advantages, first, zero-centered output: unlike the sigmoid function, the tanh function outputs values in the range of $[-1, 1]$, which helps to ensure that the activations are centered around zero. This property can lead to faster convergence during training, as it mitigates issues related to bias in the activations. Second, gradient behavior: the tanh function has a steeper gradient than the sigmoid function, particularly for inputs close to zero. This characteristic allows for more effective back propagation, leading to better learning, especially in deeper networks. Ultimately, nonlinearity: by introducing nonlinearity, the tanh function enables the model to learn complex patterns and relationships within the data, which is essential. And for each x_i, there is a gate neuron (represented by the σ node in the graph) that controls the contribution of features computed from x_i to the overall output of the cell.

E. FFM

To effectively fuse pixel-level and super-pixel-level features, we have designed an FFM based on global channel attention. This module leverages the advantages of attention mechanisms, particularly for HSI-CD tasks. The global channel attention mechanism enables the model to selectively focus on key features within the rich spectro-spatial information of HSI data. By emphasizing these important elements, the model enhances its ability to extract relevant information during the classification process. This targeted focus not only improves the accuracy of predictions but also significantly boosts the overall performance of the network. By incorporating attention mechanisms, the representation of the data becomes more dynamic, ensuring that the most significant features are highlighted, thereby facilitating better differentiation of changes in the observed scenes.

For the features T_1 and T_2 extracted from MSGM and SRGM with shape $C \times H \times W$, adopt an adaptive method based on the global spatial attention mechanism for fusion. Fig. 6 shows the structure of FFM. The specific method can be described as

$$\boldsymbol{L}_1(X) = \mathbf{Conv}_2(\delta(\mathbf{Conv}_1(\mathbf{GAP}(X_1 \oplus X_2)))) \quad (15)$$

$$\boldsymbol{L}_2(X) = \mathbf{Conv}_2(\delta(\mathbf{Conv}_1(X_1 \oplus X_2))) \quad (16)$$

$$\boldsymbol{L}_3(X) = \mathbf{Conv}_2(\delta(\mathbf{Conv}_1(\mathbf{GMP}(X_1 \oplus X_2)))) \quad (17)$$

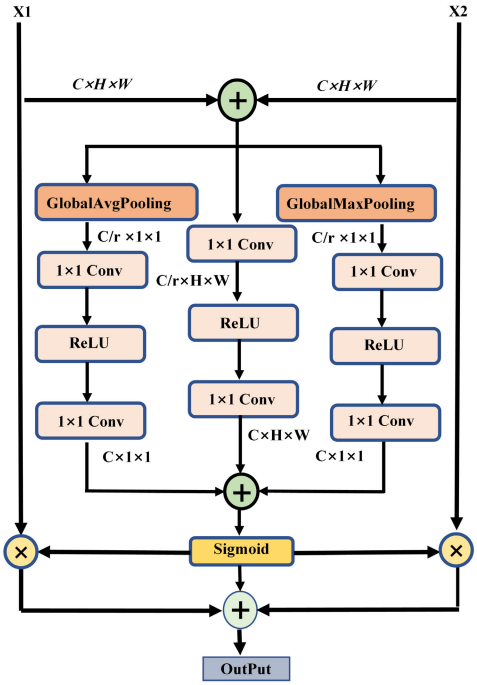


Fig. 6. Overall architecture of FFM.

where Conv_1 convolution reduces the number of input feature channels to the original $(1/r)$, in which δ represents the ReLU activation function, and restores the number of channels to the original through Conv_2 convolution. The number of incoming channels is the same, and r is the "channel scale ratio."

The calculated weight value is used to perform attention operations on the input feature to obtain the output. The formula is as follows:

$$\boldsymbol{X}_{\text{out}} = (X_1 \oplus X_2) \otimes (\boldsymbol{\delta}(\boldsymbol{L}_1(X) \oplus \boldsymbol{L}_2(X) \oplus \boldsymbol{L}_3(X)) \quad (18)$$

where δ represents the Sigmoid activation function, and the scales of the three branches are not the same; it is kept the same size as the input feature, so the addition of the two needs to be broadcast; and $\otimes$ represents the two feature map counterparts primes multiplied.

F. Loss Function

After performing the tail mapping layer with convolution, ReLU, and linear operations on F_g, the change map $\hat{Y}$ can be predicted using Softmax. For our method, naive binary cross-entropy is used as the loss function to calculate the difference between the predicted change map and the GT map

$$\text{Loss} = -\sum_{i=1}^{N} \left(Y_i \log \hat{Y}_i + (1 - Y_i) \log(1 - \hat{Y}_i)\right) \quad (19)$$

where N is the total number of samples, Y_i is the ith element of the GT map, and $\hat{Y}_i$ is the ith element of the predicted change map. And use the Adam optimizer to train the network parameters to continuously reduce loss until the network training epochs reach the maximum.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.

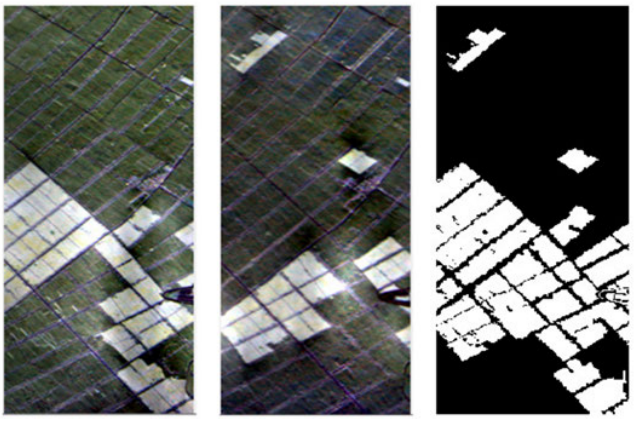


Fig. 7. China dataset. (From left to right) Image acquired on May 3, 2006, image acquired on April 23, 2007, and GT.

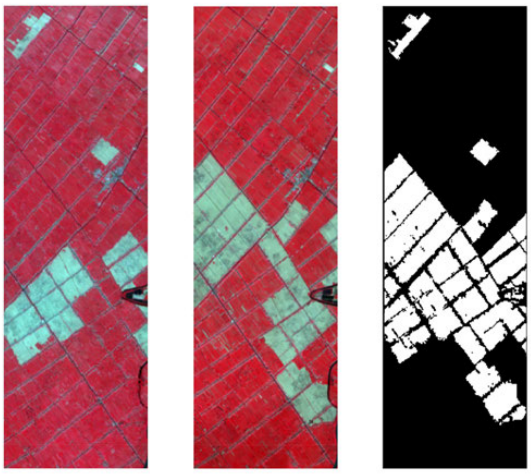


Fig. 8. Farmland dataset. (From left to right) Image acquired on May 3, 2006, image acquired on April 23, 2007, and GT.

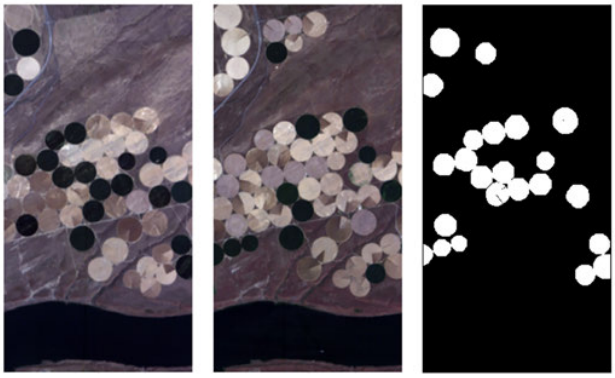


Fig. 9. Hermiston dataset. (From left to right) Image acquired on May 1, 2004, image acquired on May 8, 2007, and GT.

IV. Experiments and Analysis

A. Experimental Settings

Typical sampling rates used in most research are designed to test the effectiveness and excellent detection performance of the framework.

In this article, three HS CD datasets are applied to evaluate the performance of the SRGMGN method. Each dataset consists of three images, including two multitemporal HSIs and a GT map. The GT map composed of changed and unchanged classes is a changed difference image that reflects the real change information of ground objects.

The China dataset1 was obtained on May 3, 2006, and April 23, 2007, as shown in Fig. 7, using the Earth Observing-1 (EO-1) Hyperion HS sensor, respectively. It covers an area of farmland near the city of Yancheng, Jiangsu, China. For this dataset, the sizes of the two HSIs are both 420 × 140 pixels with a spatial resolution of 30 m. This dataset is characterized by 242 bands covering the spectral range of 0.4–2.5 μm. After removing the noise and water absorption bands, a total of 154 spectral bands are used for CD. It can be clearly seen from the GT image that the major change is in the farmland.

The Farmland dataset was captured by the Hyperion sensor on the EO-1 satellite on May 3, 2006, and April 23, 2007, in Yancheng. The pseudo-color images of bands 33, 22, and 11 and the corresponding GT are shown in Fig. 8. Each image has 155 bands with a size of 450 × 140 pixels. In total, the dataset contains 18 277 changed pixels and 44 723 unchanged pixels, which reflected the changes in the farmland due to crop rotation.

The Hermiston dataset was collected by Hyperion sensor on May 1, 2004, and May 8, 2007, in the Hermiston City area (OR, USA). The pseudo-color images of bands 33, 22, and 11 and the corresponding GT are shown in Fig. 9. Each image has 242 bands with a size of 390 × 200 pixels, and the spatial resolution was 30 m. In total, the dataset contains 9986 changed pixels and 68 014 unchanged pixels, which reflected the variations in the circular farmland area.

The proposed method is implemented using the PyTorch platform and conducted on NVIDIA GTX 3090 GPU. In the network training process, Adam is employed to optimize the networks with a learning rate of 0.0005. The number of training epochs is 300. The segmentation scales of the China, Farmland, and Hermiston datasets are set to 150, 300, and 400, and the corresponding initial numbers of superpixels are 420, 208, and 190, respectively. For China dataset, randomly selected 0.5% for training, for Farmland dataset, randomly selected 1.5% for training, for Hermiston dataset, randomly selected 10% for training, and for the three datasets, all of the rest are for testing.

Use overall accuracy (OA), Kappa coefficient, and precision (P) to quantitatively assess the detection precision of each detection method. The higher the OA, the more accurate the detection for each class. The higher the Kappa, the more consistent the detection result map is with the GT map. The higher the precision, there is a greater proportion of real change. The OA, Kappa coefficients, and precision (P) are calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{20}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - \mathrm{Pre}}{1 - \mathrm{Pre}} \tag{21}$$

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{22}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.

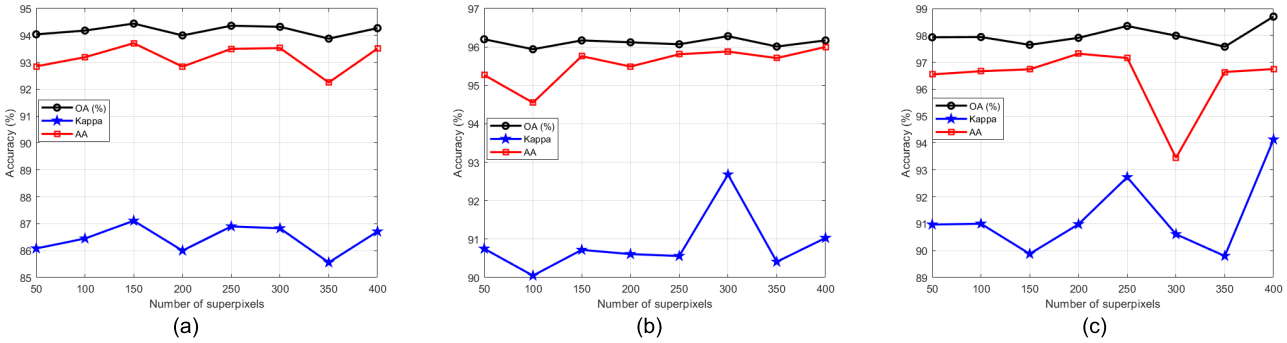


Fig. 10. Performance of the proposed method with different numbers of superpixels on (a) China, (b) Farmland, and (c) Hermiston datasets.

where TP, TN, FP, and FN stand for true positive, true negative, false positive, and false negative, respectively. Among them, TP represents the number of correctly predicted changed pixels, TN represents the number of correctly predicted unchanged pixels, FP represents the number of wrongly predicted changed pixels, and FN represents the number of wrongly predicted unchanged pixels. The intermediate value of Pre is calculated as follows:

$$\text{Pre} = \frac{(\text{TP}+\text{FP})(\text{TP}+\text{FN})+(\text{TN}+\text{FP})(\text{TN}+\text{FN})}{(\text{TP}+\text{TN}+\text{FP}+\text{FN})^2}. \quad (23)$$

B. Parameter Setting

1) Number of Superpixel: The proposed method adopts the SLIC algorithm to segment the whole image into superpixels. Since there exist differences in the size of HSI and the scale of ground objects, the optimal numbers of superpixels are different on different datasets. To find the superpixel number that makes the method achieve the best results, experiments were conducted on all the datasets. The step size of the superpixel number was set as 50, and the experimental results of three datasets are shown in Fig. 10. The performance of the method is satisfactory with different numbers of superpixel. According to the values of OA, Kappa, and AA, we choose the numbers of superpixels as 150, 300, and 400 on the China, Farmland, and Hermiston datasets to make the network obtain the best results, respectively.

C. Experimental Result

The sampling rates of all three datasets are set to 0.5%, 1.5%, and 10% under the same experimental conditions and compared with a series of recently published state-of-the-art methods in both subjective and objective aspects. The comparison methods include: D2AGCN [59], CSDBF [60], SST-Former [46], ML-EDAN [30], SATNET [63], and DBS3TAN [64].

For the D2AGCN Qu et al. [59], GCN is introduced to the HSI change detection task for the first time, which realizes the convolution operation in a non-Euclidean space and effectively preserves the class boundary.

For the CSDBF Wang et al. [60], in the superpixel-level feature extraction branch, introduce the idea of GATs into the HSI-CD field for the first time.

The SST-Former Wang, Hong, et al. [46], also employs the transformer-based structure for HSI-CD tasks. Accordingly, it is reasonable for us to select the model to compare the HSI CD performances with our proposed DBS3TAN.

For the ML-EDAN Qu et al. [30], the model is composed of two parallel architectures with multilevel encoder–decoder layers and LSTM network.

For the SATNET Zhang et al. [63], SATNet includes three components: SETrans feature extraction mod- ule, transformer-based correlation representation module, and detection module.

The DBS3TAN Zhang et al. [64], considers the advantages of CNN and transformers, combine them in two branches for spatial and spectral feature extraction, respectively. Furthermore, improve the standard transformer structure, which not only focuses on the sequence attributes of spectral signatures commonly, but also mines the spatial features from the relationship between adjacent pixels.

The quantitative metrics of the detection results are compared in Table I. The proposed SRGMGN achieves the optimal detection results, on the China dataset, CSDBF achieves suboptimal results, and on the Farmland and Hermiston datasets, SST-Former achieves suboptimal results. For the China dataset, the OA of methods has reached a relatively satisfactory level, which basically meets the needs of CD, but Kappa does not have such a high accuracy. Notably, the proposed method outperformed others, achieving the highest precision (P) values, which indicates a greater reliability in detecting changes. For the Farmland dataset, only SRGMGN can obtain OA above 0.96 and Kappa above 0.92, compared with other methods, which demonstrates greater consistency with the GT map and better detection result. And the SRGMGN has the highest P, which embodies the powerful ability of the proposed method to minimize false positives. For the Hermiston dataset, some methods can reach OA values of 0.92, but the difference between OA and Kappa is too much, suggesting that these methods may be strongly affected by class imbalance factor. The proposed method, in contrast, achieves lower difference, which suggests that it is more adept at accurately identifying minority classes. This improved precision reflects a more balanced performance across different categories, making the approach particularly effective for datasets with class imbalance issues. And the SRGMGN also achieves the highest P value. This indicates that the proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE PERFORMANCE OF DIFFERENT DETECTION METHODS IN TERMS OF OA AND KAPPA ON THREE DATASETS

Datasets	Metric	D2AGCN	CSDBF	SST-Former	ML-EDAN	SATNET	DBS3TAN	SRGMGN
China	OA (%)	86.06	92.58	92.87	91.98	92.21	90.14	**94.44**
	Kappa	0.7172	0.8320	0.8394	0.8245	0.8178	0.7876	**0.8711**
	P (%)	76.87	83.85	82.96	82.58	83.47	81.04	**87.73**
Farmland	OA (%)	87.40	95.58	95.26	94.44	93.56	91.89	**96.28**
	Kappa	0.7882	0.9125	0.9096	0.8982	0.9120	0.8276	**0.9268**
	P (%)	76.21	86.95	87.21	85.83	85.43	82.39	**90.41**
Hermiston	OA (%)	92.72	94.26	97.15	93.42	95.61	95.69	**98.70**
	Kappa	0.7737	0.8425	0.8698	0.8374	0.8265	0.8376	**0.9413**
	P (%)	83.45	86.74	91.47	84.76	89.37	88.79	**93.21**

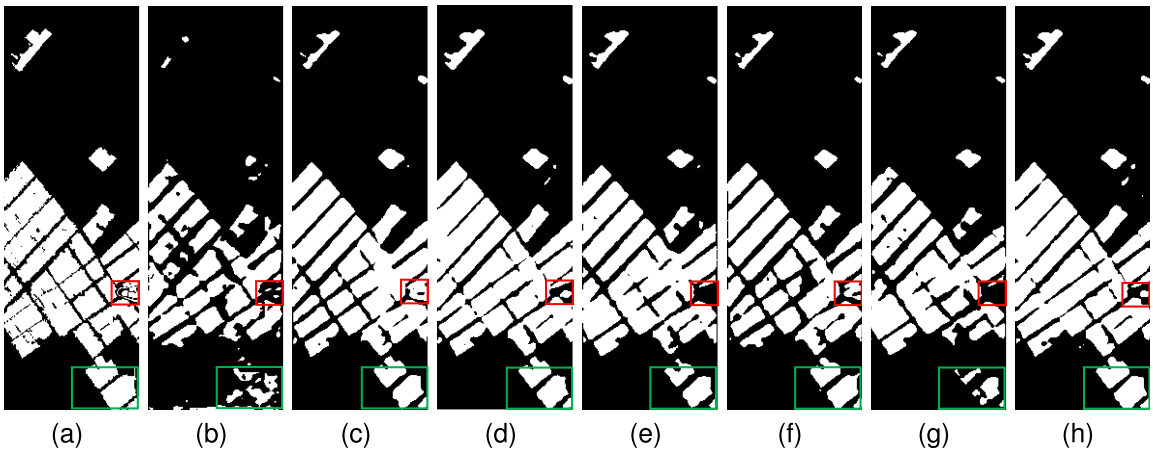


Fig. 11. CD maps on the China dataset. (a) GT. (b) D2AGCN. (c) CSDBF. (d) SST-Former. (e) ML-EDAN. (f) SATNET. (g) DBS3TAN. (h) Proposed.

method has the highest accuracy in predicting the positive sample results.

1) Experiment on the China Dataset: Fig. 11 shows a visual comparison of each method on the China dataset. It can be clearly observed that D2AGCN and DBS3TAN are relatively deficient in detecting scattered points and blocks, and CSDBF and SST-Former can detect most changes, but the results are not accuracy at the edges and small regions. ML-EDAN and SATNET have more miss alarms in the red rectangular box, and in green rectangular box, D2AGCN and DBS3TAN miss a lot of changes, and SRGMGN has the highest consistency with GT and has excellent detection performance.

2) Experiment on the Farmland Dataset: Fig. 12 shows a visual comparison of each method on the Farmland dataset. The D2AGCN and DBS3TAN are deficient in detecting scattered points and blocks, especially the D2AGCN, CSDBF, SST-Former, ML-EDAN, and SATNET are not capable of recognizing the unchanged information, resulting in noise in the background. And in the red rectangular box, in addition to the proposed methods, there are errors in detection information, while SRGMGN can achieve a superior result for detecting changed edges and other methods are weak in capturing details and have more false alarms at edges.

3) Experiment on the Hermiston Dataset: Fig. 13 shows a visual comparison of each method on the Hermiston dataset. The overall detection results of all methods on the Hermiston dataset are relatively accurate, but some of them are not capable of recognizing the unchanged information, resulting in noise in the background. In the red rectangular box and the green rectangular box, the other methods almost miss some changes. It can be seen that SRGMGN can detect most of the changed areas more accurately.

D. Ablation Experiments and Discussion

1) Ablation Experiments: The proposed SRGMGN consists of three parts: SRGM, MSGM, and FFM. To evaluate the effectiveness of each module, ablation experiments were designed and conducted on three datasets. And in order to better perform the ablation study, more detailed ablation experiments on the MSGM module are carried out. The experimental results are shown in Tables II–IV.

To fully evaluate the proposed SRGMGN, the experiments investigate the performance in terms of OA and Kappa by adding different modules in a stepwise fashion. The extensive ablation experiments are conducted on three datasets to verify the effectiveness of these components in the proposed SRGMGN for HSI-CD. As shown in Tables II–IV, through the analysis of the first three rows of the table, it is not difficult to find that the effect of a single convolution or graph convolution is not as good as the combination of graph convolution and convolution, which proves that the pixel-level features extracted by the convolution supplement the superpixel features extracted by the graph convolution, and the S-branch has worst performance; there are two reasons, first, limited feature expressiveness: pure GCNs may struggle to capture higher order features or more complex contextual relationships, which are essential for effectively identifying changes. Second, insufficient information propagation: GCNs primarily rely on information from neighboring nodes. In CD scenarios, changes may involve nodes that are farther apart,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.

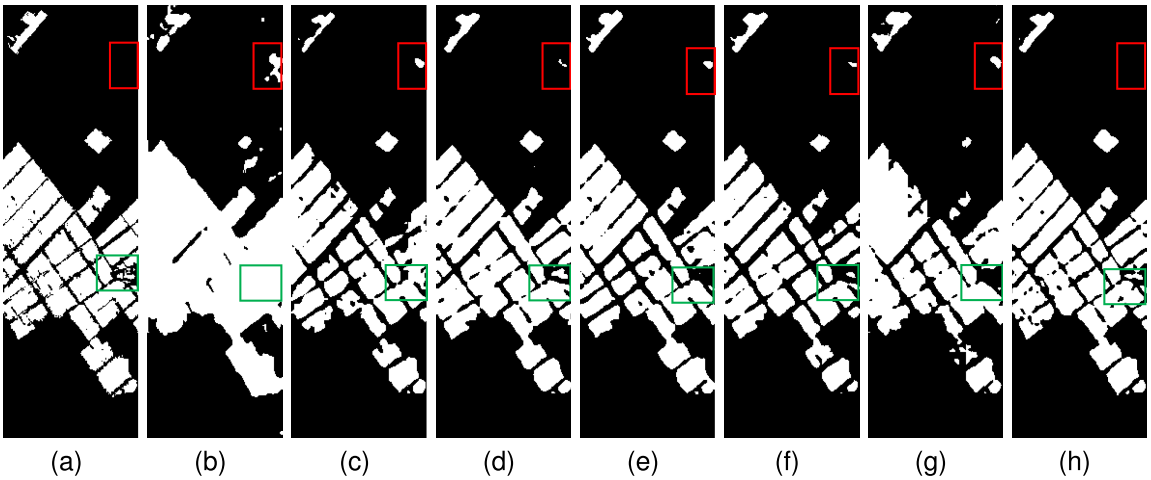


Fig. 12. CD maps on the Farmland dataset. (a) GT. (b) D2AGCN. (c) CSDBF. (d) SST-Former. (e) ML-EDAN. (f) SATNET. (g) DBS3TAN. (h) Proposed.

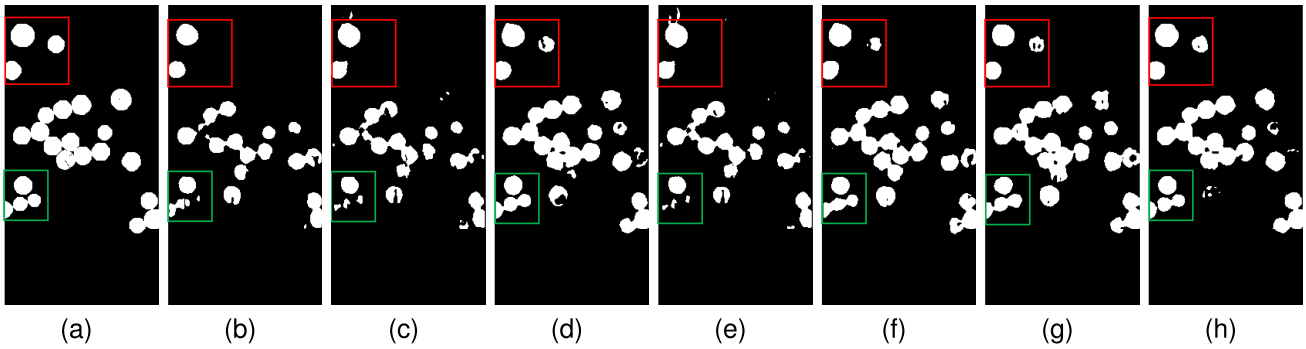


Fig. 13. CD maps on the Hermiston dataset. (a) GT. (b) D2AGCN. (c) CSDBF. (d) SST-Former. (e) ML-EDAN. (f) SATNET. (g) DBS3TAN. (h) Proposed.

TABLE II

ABLATION ANALYSIS OF THE PROPOSED MODEL WITH COMBINATION OF DIFFERENT MODULES ON CHINA DATASET

S-Branch	P-Branch	FFM	MSGM	MSGM(single CNN)	MSGM(No GCF)	SRGM	SRGM(Plain GCN)	OA(%)	Kappa
✓						✓		76.67	0.4502
	✓		✓					92.17	0.8560
✓	✓		✓			✓		93.79	0.8573
✓	✓	✓	✓			✓		94.44	0.8711
✓	✓	✓	✓				✓	93.32	0.8680
✓	✓	✓		✓		✓		94.08	0.8621
✓	✓	✓			✓	✓		93.85	0.8621

TABLE III

ABLATION ANALYSIS OF THE PROPOSED MODEL WITH COMBINATION OF DIFFERENT MODULES ON FARMLAND DATASET

S-Branch	P-Branch	FFM	MSGM	MSGM(single CNN)	MSGM(No GCF)	SRGM	SRGM(Plain GCN)	OA(%)	Kappa
✓						✓		71.60	0.4306
	✓		✓					94.41	0.8991
✓	✓		✓			✓		95.12	0.9071
✓	✓	✓	✓			✓		96.28	0.9268
✓	✓	✓	✓				✓	95.15	0.9077
✓	✓	✓		✓		✓		95.81	0.8979
✓	✓	✓			✓	✓		95.44	0.9101

leading to inadequate local information to represent global changes. By comparing the data of the third row and the fourth row, it can be found that the FFM module proposed better integrates the pixel-level features with the superpixel features and improves the accuracy of OA by nearly one percentage point or more. By comparing the data in the fourth and fifth rows of the table, it can be seen that the use of enhanced residual connections can effectively solve the problem of graph node information convergence, and by comparing the five, six, and seven rows in the table, it can also find that the multiscale change characteristics are due to the single-scale change characteristics, and the features guided by the gating mechanism are also more conducive to CD.

2) Discussion on Sampling Rates: To assess the performance of the proposed model when trained with limited samples, selecting 0.1%, 0.5%, 1%, 1.5%, 5%, and 10% as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ABLATION ANALYSIS OF THE PROPOSED MODEL WITH COMBINATION OF DIFFERENT MODULES ON HERMISTON DATASET

S-Branch	P-Branch	FFM	MSGM	MSGM(single CNN)	MSGM(No GCF)	SRGM	SRGM(Plain GCN)	OA(%)	Kappa
✓						✓		87.29	0.6916
	✓		✓					96.17	0.8472
✓	✓		✓			✓		96.40	0.9150
✓	✓	✓	✓			✓		98.70	0.9413
✓	✓	✓	✓				✓	97.89	0.9090
✓	✓	✓		✓		✓		94.65	0.8134
✓	✓	✓			✓	✓		98.01	0.9224

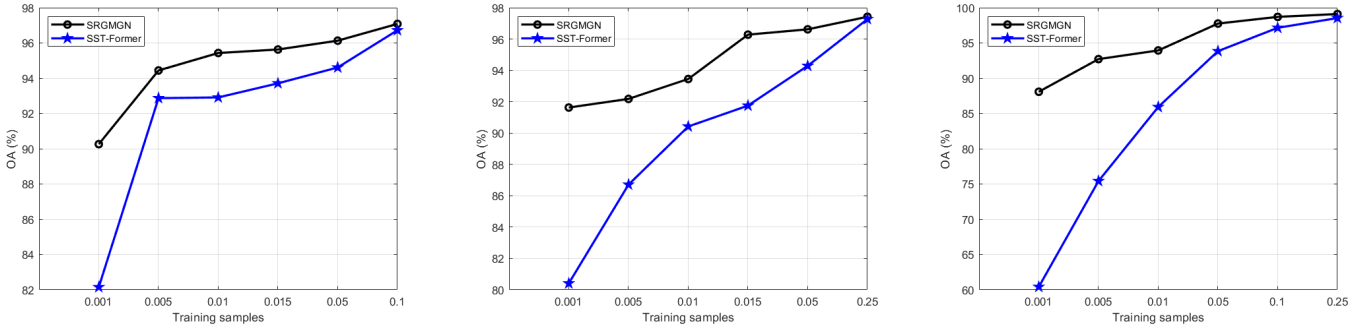


Fig. 14. OA obtained by different models with different ratios of training samples for (a) China dataset, (b) Farmland dataset, and (c) Hermiston dataset.

new training sets on the China dataset. Selecting 0.1%, 0.5%, 1%, 1.5%, 5%, and 10% as new training sets on the Farmland dataset. Selecting 0.1%, 0.5%, 1%, 5%, 10%, and 25% as new training sets on the Farmland dataset. The results are illustrated in Fig. 14. The proposed model also has great detection performance in the face of limited training samples and choose one method to compare; it can be seen that the proposed method is better.

V. CONCLUSION

In this article, a framework (SRGMGN), which includes two subnetworks of decomposed MSGM and strengthened residual graph convolution to extract the multilevel information of HSI. Specifically, the decomposed multiscale gated guided CNNs aim to comprehensively capture pixel-level features with different kernel sizes, and a GCF unit is used to fuse the multiscale pixel-level features. The strengthened residual graph convolution (SRGM) aggregates the change information to avoid the graph node information from becoming consistent and an FFM is adopted to fuse the features of two subnetworks. The SRGMGN makes full use of multiscale convolution and graph features, which is conducive to the learning of multilevel contextual semantic features. The experimental results on three HSI datasets show that the SRGMGN has a better performance than a few state-of-the-art methods. In future work, our work will focus on optimizing the network structure and promoting deeper integration of features to improve model training efficiency and accuracy.

REFERENCES

[1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag. (replaces Newsletter)*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[2] G. Lu and B. Fei, "Medical hyperspectral imaging: A review," *J. Biomed. Opt.*, vol. 19, no. 1, Jan. 2014, Art. no. 010901.

[3] X. Wei, W. Li, M. Zhang, and Q. Li, "Medical hyperspectral image classification based on end-to-end fusion deep neural network," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 11, pp. 4481–4492, Nov. 2019.

[4] H. Huang, R. Wu, Y. Li, and C. Peng, "Self-supervised transfer learning based on domain adaptation for benign-malignant lung nodule classification on thoracic CT," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 8, pp. 3860–3871, Aug. 2022.

[5] D. Mahapatra, Z. Ge, and M. Reyes, "Self-supervised generalized zero shot learning for medical image classification using novel interpretable saliency maps," *IEEE Trans. Med. Imag.*, vol. 41, no. 9, pp. 2443–2456, Sep. 2022.

[6] X. Yang, Y. Ye, X. Li, R. Y. K. Lau, X. Zhang, and X. Huang, "Hyperspectral image classification with deep learning models," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5408–5423, Sep. 2018.

[7] W. Sun, G. Yang, J. Peng, and Q. Du, "Lateral-slice sparse tensor robust principal component analysis for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 1, pp. 107–111, Jan. 2020.

[8] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[9] H. Luo, C. Liu, C. Wu, and X. Guo, "Urban change detection based on Dempster–Shafer theory for multitemporal very high-resolution imagery," *Remote Sens.*, vol. 10, no. 7, p. 980, Jun. 2018.

[10] P. Washaya, T. Balz, and B. Mohamadi, "Coherence change-detection with Sentinel-1 for natural and anthropogenic disaster monitoring in urban areas," *Remote Sens.*, vol. 10, no. 7, p. 1026, Jun. 2018.

[11] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112636.

[12] P. de Bem, O. de Carvalho Junior, R. F. Guimar aes, and R. T. Gomes, "Change detection of deforestation in the Brazilian Amazon using Landsat data and convolutional neural networks," *Remote Sens.*, vol. 12, no. 6, p. 901, Mar. 2020.

[13] K. Isaienkov, M. Yushchuk, V. Khramtsov, and O. Seliverstov, "Deep learning for regular change detection in Ukrainian forest ecosystem with Sentinel-2," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 364–376, 2021.

[14] J. Choi, B. Wang, G. Kim, and Y. Han, "Hyperspectral change detection by using IR-MAD and synthetic image fusion," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 1658–1661.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.

[15] Q. Guo, J. Zhang, and Y. Zhang, "Multitemporal hyperspectral images change detection based on joint unmixing and information coguidance strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9633–9645, Nov. 2021.

[16] F. Huang, Y. Yu, and T. Feng, "Hyperspectral remote sensing image change detection based on tensor and deep learning," *J. Vis. Commun. Image Represent.*, vol. 58, pp. 233–244, Jan. 2019.

[17] S. Liu, D. Marinelli, L. Bruzzone, and F. Bovolo, "A review of change detection in multitemporal hyperspectral images: Current techniques, applications, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 140–158, Jun. 2019.

[18] L. Ke, Y. Lin, Z. Zeng, L. Zhang, and L. Meng, "Adaptive change detection with significance test," *IEEE Access*, vol. 6, pp. 27442–27450, 2018.

[19] B. Cui, Y. Zhang, L. Yan, J. Wei, and Q. Huang, "A SAR change detection method based on the consistency of single-pixel difference and neighbourhood difference," *Remote Sens. Lett.*, vol. 10, no. 5, pp. 488–495, May 2019.

[20] S. Ertürk, "Fuzzy fusion of change vector analysis and spectral angle mapper for hyperspectral change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 5045–5048.

[21] S. Achour, M. Chikr Elmezouar, N. Taleb, K. Kpalma, and J. Ronsin, "A PCA-PD fusion method for change detection in remote sensing multi temporal images," *Geocarto Int.*, vol. 37, no. 1, pp. 196–213, Jan. 2022.

[22] J. Prendes, M. Chabert, F. Pascal, A. Giros, and J.-Y. Tourneret, "A new multivariate statistical model for change detection in images acquired by homogeneous and heterogeneous sensors," *IEEE Trans. Image Process.*, vol. 24, no. 3, pp. 799–812, Mar. 2015.

[23] A. Asokan and J. Anitha, "Change detection techniques for remote sensing applications: A survey," *Earth Sci. Inform.*, vol. 12, no. 2, pp. 143–160, Jun. 2019.

[24] H. Li, M. Gong, Q. Wang, J. Liu, and L. Su, "A multiobjective fuzzy clustering method for change detection in SAR images," *Appl. Soft Comput.*, vol. 46, pp. 767–777, Sep. 2016.

[25] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2018.

[26] T. Zhan et al., "A spectral–spatial convolution neural network with Siamese architecture for change detection," *Remote Sens.*, vol. 13, no. 5, pp. 895–912, 2021.

[27] X. Ou, L. Liu, B. Tu, G. Zhang, and Z. Xu, "A CNN framework with slow-fast band selection and feature fusion grouping for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524716.

[28] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[29] C. Shi, Z. Zhang, W. Zhang, C. Zhang, and Q. Xu, "Learning multiscale temporal–spatial–spectral features via a multipath convolutional LSTM neural network for change detection with hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5529816.

[30] J. Qu, S. Hou, W. Dong, Y. Li, and W. Xie, "A multilevel encoder–decoder attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518113.

[31] L. Gao, Z. Wang, L. Zhuang, H. Yu, B. Zhang, and J. Chanussot, "Using low-rank representation of abundance maps and nonnegative tensor factorization for hyperspectral nonlinear unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504017.

[32] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503914.

[33] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS3LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504218.

[34] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509417.

[35] L. Gao, X. Sun, X. Sun, L. Zhuang, Q. Du, and B. Zhang, "Hyperspectral anomaly detection based on chessboard topology," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505016.

[36] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2022.

[37] W. Rao, Y. Qu, L. Gao, X. Sun, Y. Wu, and B. Zhang, "Transferable network with Siamese architecture for anomaly detection in hyperspectral images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 106, Feb. 2022, Art. no. 102669.

[38] X. Sun et al., "Target detection through tree-structured encoding for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4233–4249, May 2021.

[39] X. Sun et al., "Ensemble-based information retrieval with mass estimation for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5508123.

[40] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[41] J. Bai, B. Ding, Z. Xiao, L. Jiao, H. Chen, and A. C. Regan, "Hyperspectral image classification based on deep attention graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, Art. no. 5504316.

[42] A. Song, J. Choi, Y. Han, and Y. Kim, "Change detection in hyperspectral images using recurrent 3D fully convolutional networks," *Remote Sens.*, vol. 10, no. 11, p. 1827, Nov. 2018.

[43] B. Yang, Y. Mao, L. Liu, X. Liu, Y. Ma, and J. Li, "From trained to untrained: A novel change detection framework using randomly initialized models with spatial–channel augmentation for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4402214.

[44] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Jan. 2020.

[45] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[46] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536814, doi: 10.1109/TGRS.2022.3203075.

[47] Y. Zhang et al., "A cross-modal feature aggregation and enhancement network for hyperspectral and LiDAR joint classification," *Expert Syst. Appl.*, vol. 22, Aug. 2024, Art. no. 125145.

[48] Y. Zhang et al., "Multimodal transformer network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514317, doi: 10.1109/TGRS.2023.3283508.

[49] B. Tu, Q. Ren, J. Li, Z. Cao, Y. Chen, and A. Plaza, "NCGLF2: Network combining global and local features for fusion of multisource remote sensing data," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102192, doi: 10.1016/j.inffus.2023.102192.

[50] B. Tu, X. Liao, Q. Li, Y. Peng, and A. Plaza, "Local semantic feature aggregation-based transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536115, doi: 10.1109/TGRS.2022.3201145.

[51] B. Tu, W. He, Q. Li, Y. Peng, and A. Plaza, "A new context-aware framework for defending against adversarial attacks in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505114, doi: 10.1109/TGRS.2023.3250450.

[52] B. Tu, X. Yang, B. He, Y. Chen, J. Li, and A. Plaza, "Anomaly detection in hyperspectral images using adaptive graph frequency location," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Sep. 2, 2024, doi: 10.1109/TNNLS.2024.3449573.

[53] X. Liao, B. Tu, J. Li, and A. Plaza, "Class-wise graph embedding-based active learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522813, doi: 10.1109/TGRS.2023.3309032.

[54] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-Mamba: 3D-spectral–spatial mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522813.

[55] Y. He, B. Tu, B. Liu, Y. Chen, J. Li, and A. Plaza, "Hybrid multi-scale spatial–spectral transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–18, 2024, Art. no. 5527918.

[56] L. Waikhom and R. Patgiri, "Graph neural networks: Methods, applications, and opportunities," 2021, *arXiv:2108.10733*.

[57] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, Jan. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.

[58] R. Singh, S. Bathla, and P. Meel, "State-of-the-art applications of graph convolutional neural networks," in *Proc. 6th Int. Conf. Recent Trends Comput.* Cham, Switzerland: Springer, 2021, pp. 107–115.

[59] J. Qu, Y. Xu, W. Dong, Y. Li, and Q. Du, "Dual-branch difference amplification graph convolutional network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519912.

[60] X. Wang, K. Zhao, X. Zhao, and S. Li, "CSDBF: Dual-branch framework based on temporal–spatial joint graph attention with complement strategy for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540118.

[61] F. Scarselli, A. C. Tsoi, M. Gori, and M. Hagenbuchner, "Graphical-based learning environments for pattern recognition," in *Proc. Joint IAPR Int. Workshops Stat. Techn. Pattern Recognit. (SPR) Struct. Syntactic Pattern Recognit. (SSPR)*. Cham, Switzerland: Springer, 2004, pp. 42–56.

[62] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The graph neural network model," *IEEE Trans. Neural Netw.*, vol. 20, no. 1, pp. 61–80, Jan. 2009.

[63] W. Zhang, L. Su, Y, Zhang, and X. Lu, "A spectrum-aware transformer network for change detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518612.

[64] Y. Zhang, T. Wang, C. Zhang, S. Xu, H. Gao, and C. Li, "A dual-branch Siamese spatial–spectral transformer attention network for hyperspectral image change detection," *Expert Syst. Appl.*, vol. 238, 2024, Art. no. 122125.



Shufang Xu (Member, IEEE) received the B.S. degree in applied electronic technology from Anhui Normal University, Wuhu, China, in 2002, and the master's and Ph.D. degrees in communication engineering from Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2005 and 2017, respectively.

She is currently an Associate Professor with the College of Computer and Information, Hohai University, Nanjing. In the last five years, she has participated in two research projects supported by the National Key Research and Development Program of China and one research project supported by Jiangsu Province Key Research and Development Program. Part of the research results have been applied to water resource flood disaster monitoring and operation decision-supporting system of water conservancy, which won China Industry-University-Research Cooperation Innovation Award in 2019 and 2021. Her research interests include unmanned aerial vehicle (UAV) swarm cooperation, information fusion, and image processing in remote sensing.



Xiangfei Xia received the B.E. degree in electronic information engineering from Jiangsu University of Science and Technology, Zhenjiang, China, in 2023. He is currently pursuing the M.S. degree in computer science and technology with the College of Computer Science and Software Engineering, Hohai University, Nanjing, China.

His research interests include deep learning, hyperspectral image processing, and hyperspectral change detection.



Haiwei Li received the B.E. and Ph.D. degrees in engineering from Central South University, Changsha, China, in 2009 and 2016, respectively.

He is currently a Research Associate with Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China. His research interests include remote sensing imaging mechanisms, intelligent processing of satellite-borne and UAV images, and hyperspectral applications.



Yiyan Zhang (Graduate Student Member, IEEE) received the M.S. degree (summa cum laude) from the College of Computer and Information, Hohai University, Nanjing, China, in 2022, where he is currently pursuing the Ph.D. degree with the College of Computer and Information.

His research interests include deep learning and hyperspectral image processing.



Runhua Sheng (Graduate Student Member, IEEE) received the B.E. degree in safety engineering from Zhengzhou University, Zhengzhou, China, in 2021. He is currently pursuing the M.S. degree in computer science and technology with the College of Computer Science and Software Engineering, Hohai University, Nanjing, China.

His research interests include deep learning, hyperspectral image processing, and hyperspectral image classification.



Hongmin Gao (Senior Member, IEEE) received the B.S. degree in communication engineering and the Ph.D. degree in computer application technology from Hohai University, Nanjing, China, in 2006 and 2014, respectively.

He is currently a Professor with the College of Computer and Information, Hohai University. In the last five years, he has undertaken two research projects supported by the National Natural Science Foundation of China (NSFC), one research project supported by Jiangsu Natural Science Foundation, one research project supported by the transformation of scientific and technological achievements in Jiangsu Province and participating the National Key Research and Development Program of China. Part of the research results have been applied to water resource flood disaster monitoring and operation decision-supporting system of water conservancy, which won the Second Prize of Jiangxi Province Science and Technology Advancement Reward in 2018. His research interests include deep learning, information fusion, and image processing in remote sensing.



Bing Zhang (Fellow, IEEE) received the B.S. degree in geography from Peking University, Beijing, China, in 1991, and the M.S. and Ph.D. degrees in remote sensing from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, in 1994 and 2003, respectively.

He is currently a Full Professor and the Deputy Director of the Aerospace Information Research Institute, CAS. His research interests include the development of mathematical and physical models and image processing software for the analysis of hyperspectral remote sensing data in many different areas.

Dr. Zhang was a Student Paper Competition Committee Member of IEEE International Geoscience and Remote Sensing Symposium (IGARSS) from 2015 to 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:32:17 UTC from IEEE Xplore. Restrictions apply.